

00_syllabus

September 4, 2025

1 MSE 3114: Math Methods for MSE II (AI-Augmented Materials Science) - Course Syllabus

Course Number: MSE 3114

Course Title: Math Methods for MSE II

Unofficial Course Subtitle: AI-Augmented Materials Science

Credits: 1 credit

Prerequisites: MSE 2114 (Math Methods for MSE I)

Semester: Fall 2025

Instructor: Sean Corcoran

Office Hours: Monday 2 - 3 pm, 160 Holden Hall

Email: sgc@vt.edu

1.1 Course Description & Learning Objectives

MSE 3114 introduces materials science students to the strategic integration of AI tools in research and analysis. Building upon Python programming from MSE 2114, students learn practical applications of AI tools (ChatGPT, Claude, GitHub Copilot) combined with modern data science techniques.

By the end of this course, students will be able to: 1. **Set up and configure AI tools** for materials science research 2. **Apply prompt engineering principles** for effective AI communication 3. **Utilize modern data science tools** (Polars, Streamlit, Plotly) for materials data 4. **Conduct statistical analysis** with AI assistance for interpretation 5. **Design experiments** using AI-guided optimization and DOE principles 6. **Implement machine learning models** for materials property prediction with AI assistance

1.2 Weekly Schedule & Workload

1.2.1 Weekly Schedule (15-week semester)

Week	Topic	Content	Assignment (due date)
1-2	AI Tools Setup & Basic Analysis	Environment setup, AI tools, stress-strain analysis	HW 1 (9/5/25)

Week	Topic	Content	Assignment (due date)
3-4	Prompt Engineering & Alloy Optimization	Prompt engineering, alloy analysis case study	HW 2 (9/19/25)
5-6	Modern Data Tools Introduction	Polars, Streamlit, Plotly for materials data	HW 3 (10/3/25)
7-8	Basic Statistical Analysis with AI	Descriptive/inferential statistics, assumption checking	HW 4 (10/16/25)
9-10	Basic Experimental Design with AI	DOE principles, factorial designs, LHS	HW 5 (10/31/25)
11-12	Basic Machine Learning for Materials	ML fundamentals, property prediction, classification	HW 6 (11/14/25)
13-14	Capstone Project	Integrated project applying all course concepts	Outline (11/20/25) Draft (12/5/25)
15	Course Wrap-up & Presentations	Final written project, course reflection	Final Project (12/10/25)

1.2.2 Weekly Workload Distribution

Week Type	Hours per Week	Focus
Lesson in-class Weeks (1, 3, 5, 7, 9, 11)	2-3 hours	New concept introduction and basic practice
Assignment Weeks (2, 4, 6, 8, 10, 12) (No Class)	3-4 hours	Concept application and assignment completion
Capstone Weeks (13-15) (No Class)	4-5 hours	Project work and final preparation

1.3 Required Materials

1.3.1 Software and Tools

- **Python Environment:** Anaconda or Miniconda
- **AI Tools:** ChatGPT Plus, Claude Pro, GitHub Copilot (student access)
- **Development Environment:** Cursor/VS Code or Jupyter Notebooks

1.3.2 Required Packages

Core packages (will be installed in Lesson 1)

pandas, numpy, matplotlib, seaborn, scipy, scikit-learn
polars, streamlit, plotly, psutil, itertools, warnings

1.3.3 Course Materials

- All lesson materials provided in jupyter notebook .ipynb format
 - Sample datasets for practice and projects
 - AI prompt libraries and templates
 - Reference materials and documentation
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1.4 Grading Policy

1.4.1 Grade Distribution

- **HW Assignments (6 total):** 60% (10% each)
- **Capstone Project:** 35%
- **Participation & Engagement:** 5%

1.4.2 Grading Scale

- **A:** 90-100%
- **B+:** 86-89%
- **B:** 80-85%
- **C+:** 76-79%
- **C:** 70-75%
- **D:** 60-69%
- **F:** Below 60%

1.4.3 Assignment Grading Criteria

HW Assignments (10% each): - Technical Implementation: Code execution, AI tool usage - AI Integration: Strategic AI assistance, prompt engineering - Documentation: Code comments, AI interaction logs - Results & Analysis: Appropriate interpretation

Capstone Project (35%): - Technical Report (40%): Analysis, methodology, AI integration - Interactive Dashboard (25%): Functional Streamlit application - Code Repository (20%): Well-documented, organized code - Overall Depth of Project (15%): Well-chosen multi-faceted project

1.5 Course Policies

1.5.1 Attendance and Participation

- **Attendance:** Required for weeks 1, 3, 5, 7, 9, and 11
- **Participation:** Active engagement in discussions and AI tool demonstrations
- **Collaboration:** Encouraged for learning, but individual work for assignments

1.5.2 Academic Integrity

- **AI Tool Usage:** Required and encouraged throughout the course
- **Code Attribution:** Document all AI-generated code and insights
- **Original Work:** Final submissions must represent your understanding and implementation
- **Citation:** Properly cite AI tools and external resources used

1.5.3 Assignment Submission

- **Due Dates:** All assignments due by 11:59 PM on specified dates
- **Late Policy:** 10% deduction per day after 48 hour grace period
- **Format:** Submit via Canvas

1.5.4 AI Tool Usage Guidelines

- **Strategic Integration:** Use AI tools to enhance, not replace, your learning
 - **Documentation:** Keep logs of all AI interactions for assignments
 - **Validation:** Always verify AI-generated insights and code
 - **Ethical Use:** Follow any specific guidelines per assignment description
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1.6 Support & Assessment

1.6.1 Disability Accommodations

Students with documented disabilities should contact the Office of Disability Services to arrange appropriate accommodations. Please inform the instructor of any accommodations needed.
